WAFT: Comprehensive Feature Showcase

Summary: WAFT is a scientific learning system that studies itself and evolves through the Scientific Method.

Generated: 2026-01-11 22:44 UTC | **Ideas Extracted:** 6

What Happened

> WAFT is a scientific learning system that studies itself and evolves through the Scientific Method. The system creates printable, binder-ready documents as physical knowledge artifacts. This document demonstrates every feature we've developed.

> We decided to use OAuth2 for authentication because it provides security and flexibility. The choice was made to implement adaptive constraint enforcement rather than relying on estimates. We will use WeasyPrint for PDF generation because it provides accurate page counting. The final decision was to enable metrics collection by default for all PDF generations. We chose to support PNG conversion at 300 DPI for high-quality image output.

> We discovered that character-counting methods for page estimation are unreliable. The key insight is that real measurement beats estimation every time. We learned that adaptive algorithms can achieve exact page counts through iterative adjustment. It turns out that markdown cleaning is critical for professional output. We found that comprehensive metrics enable evolution with quality data. The important realization is that fitness evaluation drives continuous improvement.

Additional Details

> We need to implement comprehensive testing for all PDF generation features. Next step is to create documentation for the metrics system. We must create example scripts demonstrating each feature. We should build a validation suite that checks all outputs. We will implement automated quality checks for generated PDFs. We need to add support for additional styling options. We should document the evolutionary event tracking system.

> The system represents a new approach to document generation. This is a framework for creating physical knowledge artifacts. The approach combines evolutionary algorithms with constraint satisfaction. The architecture uses genetic material for styling configuration. The concept of fitness drives continuous improvement. The idea of lineage tracking enables scientific naming. The framework supports multiple output formats including PDF and PNG.

> How should we handle edge cases in page counting? What is the best approach for handling very long content? Why does the system sometimes need multiple iterations? How can we improve the fitness evaluation algorithm? What metrics are most important for evolution? Which styling options provide the best readability? How do we balance completeness with constraint satisfaction? What happens when content is too short for two pages? When should we use compact versus spacious layouts? How do we determine optimal font sizes for different content types?

Content Breakdown

Type	Count
Decisions	1
Insights	1
Actions	1
Concepts	3
Questions	0